



# Yongzheng Zhang

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**Lab:** ShanghaiTech University · Declarative Systems Lab

## EDUCATION

### ShanghaiTech University

September 2024 - Present

Master's-to-Ph.D. Student in Computer Science and Technology

Shanghai, China

**Research Interests:** Explainable and Maintainable Network Configuration Verification, Synthesis, and Repair; Formal Methods; Network Protocols; Programming Languages; and Automated Reasoning

### Hainan University

September 2017 - June 2021

Bachelor of Engineering in Network Engineering

Haikou, China

## RESEARCH INTERNSHIP

### Tsinghua University

April 2024 - September 2024

*Research Intern*

*Beijing, China (Remote)*

Worked on **ClubHeap**, a high-speed, scalable programmable packet scheduler based on the PIFO (the push-in first-out queue) framework and priority queue structure with full pipeline architecture.

- Implemented the initial design of ClubHeap, a full pipeline architecture based on PIFO and priority queue.
- Developed a software simulator to verify the hardware design and conducted experiments in Vivado.

### National University of Defense Technology

July 2021 - April 2022

*Assistant Engineer*

*Changsha, China*

Worked on **ResFuz**, a fuzzing engine for estimating worst-case resource usage by resource-usage-aware fuzzing.

- Responsible for managing heap and user-defined resources, and maintaining Shell and Coccinelle scripts.
- Engaged in scientific research and submitted artifacts for FASE'22, earning available and functional badges.

## WORK EXPERIENCE

### HUAWEI · Data Communication

February 2023 - May 2023

*Software Testing Engineer*

*Nanjing, China*

Worked on **NetEngine SD-WAN AR6700v/AR6700v-lite**, which supported resource pooling, on-demand subscription and other cloud-native technologies to build a secure and reliable multi-tenant operation solution.

- Responsible for automated testing of layer-2 and layer-3 network protocols (control-plane protocols).
- Developed automated testing scripts using the Pytest framework, and implemented general check methods.

## PUBLICATIONS / ACCEPTED PAPERS

Amirmohammad Nazari, **Yongzheng Zhang**, Mukund Raghothaman, and Haoxian Chen. "Localized Explanations for Automatically Synthesized Network Configurations." In Proceedings of the 23rd ACM Workshop on Hot Topics in Networks. 2024.

**Yongzheng Zhang**, Haoxian Chen, Amirmohammad Nazari, and Mukund Raghothaman. "Interpretable Network Verification via Subspecifications." In Proceedings of the 9th Asia-Pacific Workshop on Networking. 2025.

**Yongzheng Zhang**, Yaxuan Lin, Haoxian Chen, Ruize Ma, Amirmohammad Nazari, Mukund Raghothaman, and Peng Zhang. "Explainable Network Verification via Localized Subspecification." In Proceedings of the ACM SIGCOMM 2026 Conference. 2026. **Accepted.**

**Yongzheng Zhang** and Haoxian Chen. "Characterizing Network Configuration Repair Spaces with Localized Subspecifications." In Proceedings of the 10th Asia-Pacific Workshop on Networking. 2026. **Accepted.**

Yaxuan Lin, **Yongzheng Zhang**, and Haoxian Chen. "Explaining Network Configurations Under Failures via Localized Subspecifications." In Proceedings of the 3rd Workshop on Formal Methods Aided Network Operation. 2026. **Accepted**.

## TECHNICAL SKILLS

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**Programming Languages:** Proficient in C/C++, familiar with Python, Java, P4, Shell, and Coccinelle.

**Development Tools:** Strong knowledge of the Linux environment, familiar with Vim, GCC, GDB, and Make.

**Analytical:** Passionate about programming and research, skilled in identifying, analyzing, and solving problems.